

# Anchor Ball Regression Model for large-scale 3D skull landmark detection

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## ARTICLE INFO

Communicated by P. Yan

Dataset link: [https://github.com/ithet1007/mmld\\_code](https://github.com/ithet1007/mmld_code)

### Keywords:

Encoder–decoder networks  
3D skull landmark detection  
CT volumes

## ABSTRACT

Recent deep learning models have exhibited impressive performance in the area of 3D skull landmark detection, but most of them aimed to detect a fixed number of landmarks. This paper focuses on automatically detecting an arbitrary number of landmarks on CT volumes, which meets the real clinical needs. To achieve robust performance for detecting arbitrary molar landmarks, we propose a novel 3D landmark detection model named Anchor Ball Regression Model (ABRM), which combines landmark detection and landmark classification losses for network training. For landmark detection, a novel landmark regression loss is proposed by predicting offsets to anchor balls instead of directly predicting landmarks. For landmark classification, an online hard negative mining loss is used to reduce absent landmarks' learning errors, and a small regularization constraint loss is performed for voxels outside the anchor balls. The network backbone of ABRM is obtained by manually pruning popular 3D-CNNs. We also present an available large-scale benchmark dataset in this paper, which, to the best of our knowledge, is the largest dataset for 3D skull landmark detection. The dataset comprises of 658 CT volumes, with 14 landmarks labeled by two junior and one senior doctors. The ABRM presents a good robust performance and outperforms other models on this dataset. The codes and dataset are accessible at [https://github.com/ithet1007/mmld\\_code](https://github.com/ithet1007/mmld_code).

## 1. Introduction

Identifying the locations of landmarks on the surface of bones or teeth in three-dimensional (3D) skull slices is an essential pre-processing step for clinical measurements and surgery planning. The accuracy of landmark detection is critical for the subsequent clinical steps. However, manual landmarking by surgeons can be incredibly time-consuming. Detecting landmarks on a skull volume is particularly challenging since the landmark location should be identified synchronously on intersecting, sagittal, and coronal slices. 3D landmark detection on skull slices is an important prerequisite for diverse biological and medical applications, including clinical dentistry, comparative anatomy, and anthropology. Consequently, there is a need to develop an automatic landmark detection method with high accuracy.

Various traditional methods such as knowledge-based algorithms [1], regression forests [2,3], and related projections [4] have been employed to address this issue in the past. However, the recent advent of automatic landmark detection on 3D skull slices in the deep learning community has gained much attention. Most existing methods utilize encoder–decoder networks to process the skull slices. To enhance the accuracy of automatic landmark detection, techniques like deep reinforcement learning [5,6], two-stage learning [7–12], and multi-task

learning (MTL) [13–16] have been utilized. Nevertheless, it is worth noting that most of these methods have primarily focused on small and private datasets, and their generalization performance on larger datasets remains questionable.

In this paper, we have identified two limitations that hinder the domain of 3D skull landmark detection. Firstly, the majority of studies collected data only during pre or postoperative stages. However, in a real-world clinical environment, the model must be robust enough to meet clinical demands with diverse data. Conducting a clinical landmarking evaluation is necessary in both pre and postoperative stages. Unfortunately, most models only focus on identifying a fixed number of landmarks on standard CT or CBCT volumes. As an illustration, Lu et al. [25] excluded subjects with missing incisors since the maxilla and mandible have four landmarks located on the incisors. Nevertheless, in the clinical environment, landmarking on CT volumes containing missing landmarks is common as tissue or organ damage or absence frequently occurs. Therefore, detecting a fixed number of landmarks does not fully meet clinical needs. Ultimately, the clinical need entails the 3D landmark detection model to recognize the landmarks' existence as well as locate them simultaneously. Currently, most methods using heatmap or offset regression exhibit low levels of robustness, causing

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<https://doi.org/10.1016/j.neucom.2023.127051>

Received 23 July 2023; Received in revised form 7 October 2023; Accepted 14 November 2023

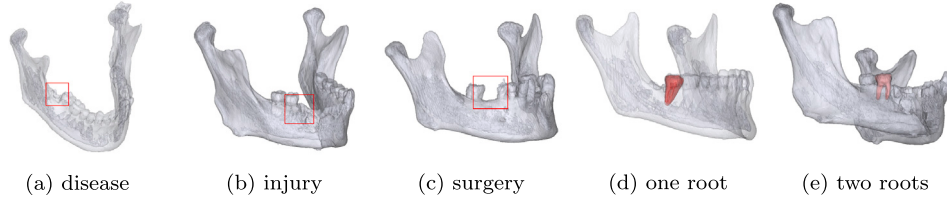
Available online 17 November 2023

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**Table 1**

Recent advanced works. The metrics VED, MRE, SDR, and RMSE represent the voxel Euclidean distance, mean radial error, successful detection rate, and root mean square error, respectively. CBCT and CT refer to the data types of cone-beam computed tomography and spiral computed tomography, respectively.

| References | Years | Number of landmarks | Available | Single task | Metric  | Data type | Data size  |
|------------|-------|---------------------|-----------|-------------|---------|-----------|------------|
| [16]       | 2018  | 9                   | ✗         | ✗           | VED     | CBCT      | 50         |
| [17]       | 2018  | 33                  | ✗         | ✓           | VED     | CBCT      | 39         |
| [18]       | 2019  | 7                   | ✗         | ✓           | MRE     | CT        | 27         |
| [13]       | 2020  | 15                  | ✗         | ✗           | MRE     | CBCT&CT   | 77&30      |
| [19]       | 2020  | 5                   | ✗         | ✓           | MRE&SDR | CBCT      | 102        |
| [14]       | 2020  | 64                  | ✗         | ✗           | RMSE    | CBCT&CT   | 77&63      |
| [8]        | 2021  | 5                   | ✗         | ✓           | MRE     | CT        | 19         |
| [20]       | 2022  | 105                 | ✗         | ✓           | RMSE    | CBCT      | 50         |
| [21]       | 2022  | 17                  | ✗         | ✓           | MRE&SDR | CBCT      | 89         |
| [22]       | 2022  | 33                  | ✗         | ✓           | MRE&SDR | CT        | 198        |
| [23]       | 2022  | 33                  | ✗         | ✓           | MRE&SDR | CBCT      | 150        |
| [24]       | 2022  | 4                   | ✗         | ✗           | VED     | CT        | 274        |
| [25]       | 2023  | 24                  | ✗         | ✓           | MRE&SDR | CBCT      | 150        |
| [26]       | 2023  | 9                   | ✗         | ✓           | RMSE    | CBCT      | 250        |
| Ours       | 2023  | 14                  | ✓         | ✓           | MRE&SDR | CT        | <b>648</b> |



**Fig. 1.** The examples with different categories of mandibles. The mandibles have a semi-hyaline appearance, while the teeth appear solid and have a red hue. Any damaged regions are indicated by red bounding boxes. Specifically, subfigures (a–c) depict teeth that have been damaged by disease, injury, and surgery, respectively. On the other hand, subfigures (d–e) illustrate teeth that have one or two roots.

non-existent landmarks to be falsely detected. Consequently, detecting an arbitrary number of landmarks on CT volumes has received relatively little attention in the literature.

Secondly, the lack of a standard benchmark dataset to evaluate the performance of automatic landmark detection models is a significant constraint. Through a review of advanced models from 2018 to 2023, listed in Table 1, it was found that these models were trained on private datasets with variable data sizes, types, evaluation metrics, and numbers of landmarks. Four models [13,14,16,24] employed multi-task learning, which cannot be compared fairly with single-task 3D landmark detection models. Moreover, the number of landmarks ranged from 4 to 105, making it challenging to assess the models' performance, given that the difficulty of 3D landmark detection varies with the number of landmarks. The evaluation metrics used in these studies were voxel Euclidean distance (VED), mean radial error (MRE), successful detection rate (SDR), and root mean square error (RMSE). Additionally, the data types used in these studies included cone-beam computed tomography (CBCT) and spiral computed tomography (CT), which led to the subdivision of the domain of 3D skull landmark detection into small sub-fields due to the differences in metrics and data types, preventing a fair comparison of these models. The dataset sizes ranged from 19 to 274, with smaller datasets evaluated using cross-validation and larger datasets using training-testing evaluation. In conclusion, developing a benchmark dataset for evaluating 3D skull landmark detection models is essential.

To address the aforementioned issues, we constructed a mandibular molar landmarking (MML) task on an available largescale dataset. The task involves detecting arbitrary landmarks, aiming to identify the anatomical locations of the crowns and roots of mandibular molars. MML presents two main challenges. Firstly, mandibular molars can be damaged by dental diseases, injury, or surgery, as illustrated in Fig. 1(a–c). Secondly, mandibular molars have varying numbers of

roots, as illustrated in Fig. 1(d–e). Encoder–decoder networks are usually insensitive to discerning the existence of landmarks. Consequently, producing false landmarks can easily become a performance bottleneck, given the diversity of CT volumes. The main contributions of this paper is as follows:

- A robust Anchor Ball Regression Model (ABRM) is proposed to improve the robustness of 3D skull landmark detection on CT images via combining landmark regression and classification losses for network training.
- By predicting the offset of the anchor balls, a novel landmark regression loss method is proposed, which offers better performance than the typical heatmap and offsetmap regression methods.
- The proposed classification loss function reduces the misclassification of existent and absence landmarks. Specifically, multiple binary cross-entropy losses are used in landmark classification to reduce the learning error of existent landmarks. An online hard-negative mining method is used to construct the learning task for absent landmarks. In addition, a regularization constraint loss with a small weight factor is used for voxels outside the anchor balls.
- A large-scale benchmark dataset consisting of 648 CT volumes is created to evaluate 3D skull landmark detection. The dataset and codes are publicly available on [https://github.com/ithet1007/mml\\_d\\_code](https://github.com/ithet1007/mml_d_code), which is the largest publicly available dataset to our knowledge.

The rest of this paper is structured as follows: in Section 2, we introduce the related work about existing 3D landmark detection methods; in Section 3, we present the proposed ABRM; in Section 4, we introduce the benchmark dataset; in Section 5, the experimental results and analysis are given; finally, we conclude our work and discuss the future study in Section 6.

## 2. Related work

### 2.1. Traditional machine learning methods

Landmark detection on CT volumes is an important prerequisite for diverse biological and medical applications, including clinical dentistry, comparative anatomy, and anthropology. Traditional methods including knowledge-based algorithm [1], regression forest [2,3], and related projection [4], were used to solve this problem. With the knowledge-based algorithm, Gupta et al. [1] first searched seed points and found empirical points through the distance vector. According to the volume of interest, landmarks were detected on the contour of the anatomical structure. Zhang et al. [2] proposed a regression voting strategy to detect landmarks by aggregating evidence from context locations. The inconsistent landmark appearance problem, caused by morphological variations among different patients, was solved by anatomical segmentation. Urschler et al. [3] attempted to combine image appearance and geometric landmark configuration into a random forest regression model. Montúfar et al. [4] first built orthogonal, coronal, and sagittal projections and trained two active shape models to locate cephalometric landmarks. Then, the related projection was used to detect the rest landmarks. Most of these traditional methods worked on small datasets with less than 30 volumes, and the generalization performance was not guaranteed in large-scale datasets.

### 2.2. Deep learning methods

With the development of deep learning in recent years, automatic landmark detection was well developed on large-scale CT volumes. Many works developed 3D-CNNs with specific algorithms to improve the accuracy of landmark detection. Kang et al. [27] developed an automatic 3D cephalometric annotation system to annotate human skulls by 3D-CNNs. Li et al. [19] proposed to replace heatmap regression with directly coordinate regression. An attention-guided mechanism was proposed to combine global landmark configuration with local high-resolution features. Ghesu et al. [5] attempted to reformulate the detection problem as a behavior learning task and used deep reinforcement learning to distinguish the target anatomical landmarks. They used a coordinate descent algorithm to optimize joint landmark predictions iteratively. Alansary et al. [6] used reinforcement learning to identify the optimal path to the landmark by interacting with CT volumes.

Several recent studies split landmark detection into several sequential sub-stage tasks to simplify the detection task. The two-stage learning is a trade-off between performance and computation. Codari et al. [7] segmented bone tissues from CT volumes and then used an intensity-based registration to detect landmarks. Palazzo et al. [8] first located landmarks in low-resolution CT scans and then refined the results at a full-resolution scale. Payer et al. [9] split the detection task into two components, where one component localized accurate but ambiguous candidate landmarks and the other improved the robustness by incorporating the spatial configuration of landmarks. Yun et al. [10] proposed a coarse-to-fine deep learning model for landmark detection on the midsagittal plane with a basic landmark annotator for 3D skull pose normalization, where the low-dimensional representations were abstracted by variational autoencoder.

Landmark detection can be performed with other medical image analysis tasks. Thus, the MTL technique has been widely used in recent years. Because most of the landmarks are located on the boundaries of bones and the spatial 3D context information is implicitly associated between voxels and landmarks, the bone segmentation and landmark detection tasks can be jointly studied. Zhang et al. [13] proposed a context-guided fully convolutional network, utilizing a displacement map to learn the spatial context information, to perform bone segmentation and landmark detection together. Lian et al. [14] proposed

a multi-task dynamic transformer network with a collaborative two-branch architecture and adaptive transformer modules to jointly regress landmark heatmaps on CT volumes. To detect bifurcation landmarks, Tan et al. [15] predicted landmark confidence maps, branch segmentation maps, and branch orientation fields in a multi-task manner. MTL is not the only way for bone segmentation and landmark detection. Torosdagli et al. [16] used a fully convolutional DenseNet for segmentation and a U-Net with a long short-term memory (LSTM) for landmark detection.

## 3. The proposed methods

### 3.1. Motivations

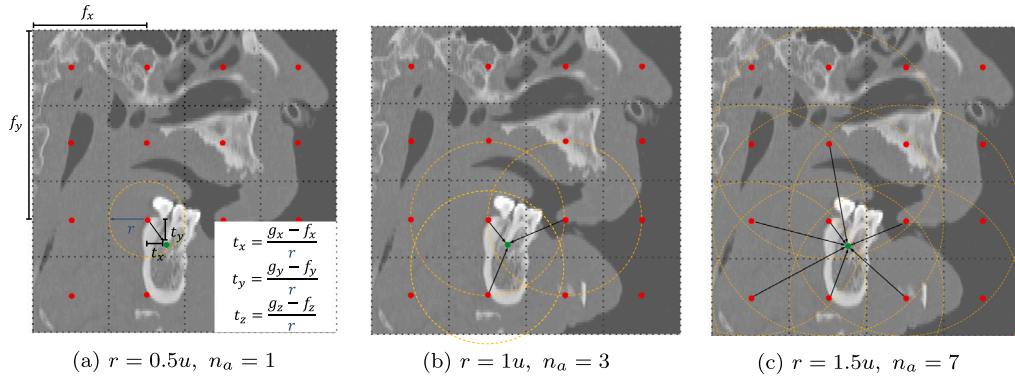
For solving the arbitrary landmark regression problem, our original idea is motivated by object detection. Object detection aims to classify an arbitrary number of objects into correct categories and to detect the bounding boxes of objects. The object detection has been well solved by You Only Look Once (YOLO) series [28–30]. YOLO predicted the bounding boxes and class probabilities in a single detection pipeline with the following important components: the anchor boxes with hand-picked priors were designed to jointly predict bounding boxes; the regression and classification losses were combined with balanced learning factors; the post-processing step of non-maximum suppression (NMS) sought out the most latent bounding boxes from candidates.

Anchor box was first proposed in Faster RCNN [31] and played an important role in telling where the objects were. Several anchor boxes deployed each grid cell with different scales and aspect ratios. The anchor boxes can be viewed as various molds for producing candidate bounding boxes. Anchor boxes most possibly predicted objects with similar scales and aspect ratios. In this paper, we attempt to involve the offsetmap regression with anchors for arbitrary landmark detection.

### 3.2. Anchor ball regression algorithm

There are two primary categories of existing landmark detection methods: First, coordinate regression-based methods, such as [19], use a fully connected prediction layer to map the input image directly to landmark coordinates. The advantages of this approach include outputting coordinate points, rapid training and inference speeds, and end-to-end differentiability during training. However, its main limitation is the absence of spatial generalization capabilities. Second, predictive Gaussian heatmap methods, such as [32], are employed to obtain landmarks by identifying the peak probability locations on the heatmap. These methods generally show improved accuracy due to their preservation of the spatial structure of image features. However, they tend to prioritize local texture information over global perception and are vulnerable to challenges such as large pose variations, severe occlusions, and complex lighting conditions, as demonstrated in [33].

In recent studies, various methods have been proposed to enhance the accuracy and robustness of 3D landmark detection models. For instance, techniques such as combining heatmaps and offset maps for pixel-wise regression voting [34], leveraging multiorder spatial correlation and multiorder channel correlation [35], as well as incorporating multiorder cross-information [36], have been developed. These advancements effectively reinforce global shape constraints and contribute to the overall performance of the model. Notably, these enhanced heatmaps are generated with all default feature points included, which does not satisfy the requirement of being resilient to missing landmarks. Consequently, our goal is to develop a robust 3D landmark detection model capable of performing anchor ball regression-based landmark detection. Inspired by YOLOV3, we downsample the CT volume into  $\beta_x \times \beta_y \times \beta_z$  grids using a 3D-CNN feature extractor. If a landmark falls within a grid cell, that grid cell becomes responsible for detecting it. Since a grid cell is mapped from a source volume region, it



**Fig. 2.** The illustration of anchor balls in the sagittal slice. The anchor balls are depicted with yellow circles. The red dot indicates the center of the grid cell, the green dot is a tooth root, and the unit of the grid cells' length is defined as  $u$ . In the subfigure (a),  $r = 0.5u$  is the radius of the anchor ball;  $f_x = 1.5u$ ,  $f_y = 2.5u$  ( $f_z$  is in the  $z$  axis and not depicted) indicate the absolute coordinate of the anchor ball;  $(t_x, t_y, t_z)$  indicates the relative coordinate of the tooth root; and  $(g_x, g_y, g_z)$  indicates the absolute coordinate of the tooth root.  $n_a = 1$  indicates that the tooth root is detected by only one anchor ball. In subfigure (b) and (c), the tooth root is detected by 3 and 7 anchor balls with  $r = 1u$  and  $r = 1.5u$ , respectively.

should detect all landmarks located in that region. Accordingly, detecting the coordinates of landmarks becomes a matter of predicting the offsets to anchor balls. This approach makes learning offset regression easier than directly detecting landmark locations.

We conclude that anchors serve two primary purposes. Firstly, they act as indicators with pre-selected priors to jointly predict candidates for ground truth. In the inference stage, the most latent prediction was selected from those candidates with largest confidences. Secondly, the anchors determine which ground truth will be detected by possessing certain properties. For object detection, the scale and aspect ratio serve as anchor properties. However, in 3D landmark detection, a landmark does not have width and height properties, so anchor boxes cannot be used directly in detecting 3D landmarks. Instead, since a landmark's location is typically related to the local features of adjacent teeth, bones or tissues, we opt to use radius  $r$  as the only anchor property in 3D landmark detection. Anchors are implemented in the form of anchor balls, which are positioned at the center of each grid cell, and the relative location to the radius  $r$  is observed to detect landmarks within the anchor balls. An example of ABRM is illustrated in Fig. 2. By deploying anchor balls of varying radii, we convert absolute coordinates to relative coordinates of landmarks. The relative coordinates are normalized to the range of  $[-1, 1]$ , allowing for cost-effective detection of landmarks by neighboring anchor balls.

### 3.3. Training losses

To learn MML task, we combine landmark regression loss and landmark classification loss during network training. The ground truth for a target landmark is represented by its absolute coordinate,  $(g_x, g_y, g_z)$ , and its corresponding category,  $c$ . Given  $L$  anchor balls with different radii, the learning target of ABRM is to predict  $(t_x^m, t_y^m, t_z^m, c)$ , where  $1 < m \leq n_a$  and  $n_a$  satisfies that

$$n_a = \sum_{l=1}^L \sum_{i,j,k} \mathbb{1} \left\{ \|(g_x, g_y, g_z) - (f_x^i, f_y^j, f_z^k)\|_2 < r_l \right\}. \quad (1)$$

$\mathbb{1}\{\cdot\}$  is the indicator function.  $1 \leq i \leq \beta_x$ ,  $1 \leq j \leq \beta_y$ ,  $1 \leq k \leq \beta_z$  are the indexes. The relative coordinates are calculated as follows:

$$\begin{cases} t_x = \frac{g_x - f_x}{r}, \\ t_y = \frac{g_y - f_y}{r}, \\ t_z = \frac{g_z - f_z}{r}. \end{cases} \quad (2)$$

Thus, landmarks are detected by adjacent anchor balls with different radius scales. The landmark regression loss is

$$J_{reg} = \sum_{m=1}^{n_a} \left\| (t_x^m, t_y^m, t_z^m) - \sigma(p_x^m, p_y^m, p_z^m) \right\|_2, \quad (3)$$

where  $(p_x^m, p_y^m, p_z^m)$  are the network prediction for detection at corresponding coordinates and  $\sigma(\cdot)$  is the Tanh activation function.

For landmark classification, we consider three situations. First, if landmarks exist, we use multiple binary cross-entropy losses to build the learning task as follows

$$J_{cl}^{(1)} = \log \xi(q_c) + \frac{1}{S-1} \sum_{s \neq c} (1 - \log \xi(q_s)), \quad (4)$$

where  $S$  is the full landmark number.  $q_c$  is the network prediction for category  $c$ .  $\xi(\cdot)$  is the Sigmoid activation function. Second, if landmarks do not exist, we apply hard negative mining to select  $H$  negative samples and build the loss function as follows

$$J_{cl}^{(2)} = \lambda_{hn} \frac{1}{H} \frac{1}{S} \sum_h \sum_s (1 - \log \xi(q_s^h)). \quad (5)$$

In this paper, we have empirically set  $\lambda_{hn} = 0.2$  and  $H = 256$ . Third, the last two cost functions penalize training errors for grid cells inside anchor balls. Since most voxels on the CT volumes will not participate in the last two loss functions, To deactivate these useless voxels, we have included a restriction term to deactivate these non-participating voxels in the entire volume, which is shown as follows

$$J_{cl}^{(3)} = \lambda_{res} \frac{1}{V} \frac{1}{S} \sum_v \sum_s (1 - \log \xi(q_s^v)), \quad (6)$$

where  $\lambda_{res}$  is set to 0.001. To sum up, the loss function for training ABRM is

$$J_{total} = J_{reg} + J_{cl}^{(1)} + J_{cl}^{(2)} + J_{cl}^{(3)}. \quad (7)$$

### 3.4. Network structures

During network training, we randomly crop a patch with dimensions of  $128 \times 128 \times 64$  from the volume to take into account the limitations of GPU memory. We use a 3D-CNN network backbone to extract features and transform the network input into grid cells with dimensions of  $32 \times 32 \times 16$ . We apply a  $1 \times 1 \times 1$  convolution as the pre-activation layer to generate  $(3 + S) * L$  final predictions for each grid cell. We obtain a network backbone by pruning popular encoder-decoder 3D-CNNs, such as VNet [37], UNet3D [38], and Residual UNet3D [39], manually. UNet is a widely used 2D encoder-decoder structure for medical image analysis. VNet is the first 3D version of UNet that was used for 3D image segmentation on MRI volumes. The most important network module of VNet is the shortcut connection from the feature maps of early stages to those of deep stages. UNet3D has similar structures to VNet but has two differences: it introduces 3D Batch Normalization [40] and deploys a kernel size of  $3 \times 3 \times 3$  instead



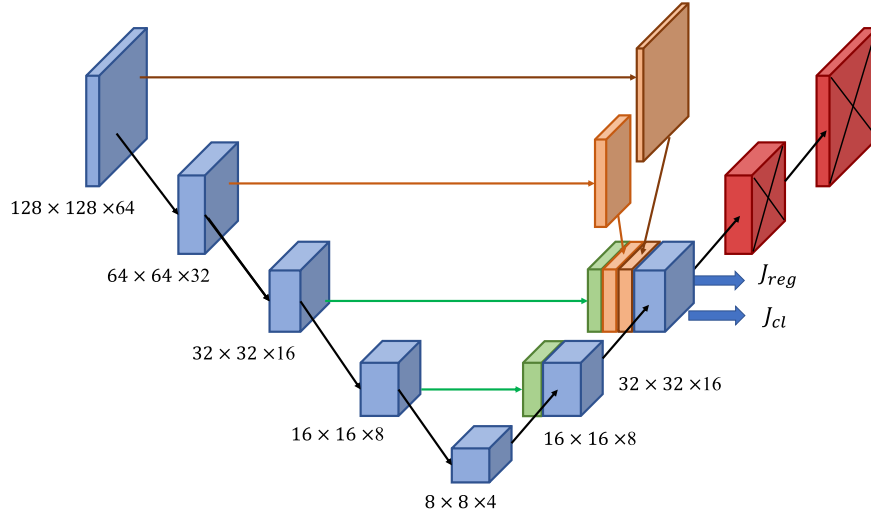


Fig. 3. The network backbone is pruned from encoder-decoder 3D-CNNs. The feature maps with red color is pruned.

of  $5 \times 5 \times 5$  in VNet. Residual UNet3D replaces the convolution blocks of UNet3D with Residual blocks. These network backbones consist of an encoder module with 3 or 4 downsample blocks and a decoder module with 3 or 4 upsample blocks. To generate grid cells of 4 times smaller than CT volumes, as illustrated in Fig. 3, we prune the last two upsample blocks. A  $1 \times 1 \times 1$  3D convolution with 2 or 4 strides is used to downsample the feature maps of early stages in orange color. Finally, the feature maps from different stages are concatenated and, a pre-activation layer is followed to generate the final network output.

### 3.5. Network inference

After the model is trained, the CT volumes on the validation and test datasets are inferred by the model. We sort the landmark classification probabilities of all voxels and select  $M$  voxels with maximum classification probabilities, where  $M$  is a fixed hyperparameter and can be set to an approximate value to  $n_a$ . The landmark candidate set is defined as  $\{(h_x^m, h_y^m, h_z^m) \mid m \in M\}$  and the landmark candidates are produced by

$$\begin{cases} h_x^m = f_x^m + r^m \cdot \sigma(p_x^m), \\ h_y^m = f_y^m + r^m \cdot \sigma(p_y^m), \\ h_z^m = f_z^m + r^m \cdot \sigma(p_z^m). \end{cases} \quad (8)$$

$(f_x^m, f_y^m, f_z^m)$  is the coordinates of  $m$ th voxels. Finally, we determine the most probable landmark by searching the minimum Euclidean distance to other landmarks as follow

$$m^* = \arg \min_{m \in M} \sum_{\substack{n \in M \\ n \neq m}} \|(h_x^m, h_y^m, h_z^m) - (h_x^n, h_y^n, h_z^n)\|_2 \quad (9)$$

## 4. Dataset and evaluation metric

The dataset was obtained from West China Hospital and comprises of 658 CT volumes (458/training, 100/validation, 100/testing). The dataset contains 384 males and 274 females, with an age range of 15 to 72 years. The CT examinations were carried out between 2014 and 2021. The CT volumes varied in scale and spacing, ranging from (0.35, 0.35, 0.35) to (0.62, 0.62, 1.05). To ensure uniformity in the dataset, we transformed the volumes to a uniform scale of  $512 \times 512 \times 256$ .

To protect the privacy of the patients, we extracted the voxels from the CT volumes and saved them in “nrrd” format. The spacing of the CT volumes was also saved, which aided in MRE calculation for evaluation purposes. In order to enhance the visibility of teeth in the CT volumes, we truncated the Hounsfield units of the voxels to [1000, 2000] and normalized them to [0, 255] using bilinear interpolation. The dicom attributes of the patients were not available in the released dataset. In the training phase, we first scan the labels of all landmarks on the CT volume and scale them accordingly to the spacing among landmarks to ensure that they meet the  $128 \times 128 \times 64$  random cropping target size, i.e., all the landmarks must be able to be included in the target size at the same time (with a default scaling of 0.6). Then, random cropping is performed, which kept that all landmarks are included in the cropped volume. During the evaluation and testing phase, we performed center cropping on the CT volume. The scaling method was the same as the scaling method for random cropping.

The dataset was labeled by two junior oral doctors and then re-evaluated by a senior oral doctor, who made necessary corrections when the average Euclidean distance between the labels was less than 1 mm. Detecting landmarks in this dataset, which contains arbitrary landmarks in CT volumes, is a challenging task for MML. The dataset includes 14 mandibular landmarks, consisting of 4 crowns, 8 roots of the second and third mandibular molars, and 2 cusps of cuspids. These landmarks are specified in Fig. 4. The second and third mandibular molars are connected to the cortical plates and the inferior alveolar canal. According to the Winter or Pell and Gregory classification system [41], the location of the third mandibular molar is significantly associated with the type of tooth impaction. Automatic detection of crown and root landmarks for second and third mandibular molars during pre-operative assessment may help to reduce postoperative complications. Impacted cuspids occur when neighboring teeth block them, so it is important to identify cuspid landmarks as well.

In Table 1, recent studies have utilized VED, MRE, SDR, and RMSE as criteria for evaluating their models. Among these, MRE and SDR have been widely used and are considered standard metrics for 2D cephalometric landmark detection [11,12,42]. Therefore, we use MRE with standard deviation, and SDR as our evaluation metrics. MRE represents the mean Euclidean distance between the predicted and ground truth values, while SDR measures the accuracy of detected landmarks within varying distances, specifically, within 2 mm, 2.5 mm,

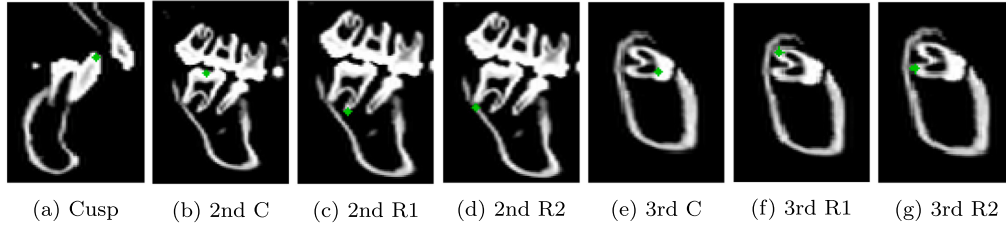


Fig. 4. Seven landmarks. “2nd” and “3rd” indicates the 2nd and 3rd molars, respectively. “C” indicates Crown and “R” indicates Root.

3 mm, and 4 mm. The formulas for these metrics are as follows:

$$\left\{ \begin{array}{l} R_i = \sqrt{\sum_{\tau \in \{x,y,z\}} ((g_{\tau}^i - f_{\tau}^i) \cdot s_{\tau}^i)^2}, \\ MRE = \frac{1}{n} \sum_{i=1}^n R_i, \\ Std = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (R_i - MRE)^2}, \\ SDR = \frac{\text{number of accurate detection}}{\text{number of detection}} \times 100\%. \end{array} \right. \quad (10)$$

where  $n$  denotes the number of CT volumes and  $R_i$  denotes the mean Euclidean distance between the ground truth landmark coordinates  $(g_x, g_y, g_z)$  and predicted landmark coordinates  $(f_x, f_y, f_z)$  of the  $i$ th volume.  $(s_x, s_y, s_z)$  indicates the spacing of the  $i$ th volume.

The main challenge faced in MML is accurately detecting an arbitrary number of landmarks. The statistical distribution of landmarks in the training set is stated in Table 2. A reliable automatic landmarking model should minimize the number of erroneously undetected landmarks. Therefore, in order to assess the robustness of landmark identification, we use recall and f1-measure as evaluation metrics for landmark classification results, as follows:

$$Recall = \frac{TP}{TP + FN} \quad (11)$$

$$F1 = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \quad (12)$$

$$\left\{ \begin{array}{l} TP : \text{existent landmarks are detected,} \\ FP : \text{nonexistent landmarks are detected,} \\ FN : \text{existent landmarks are not detected.} \end{array} \right. \quad (13)$$

## 5. Experimental results

To augment the data, we randomly zoom and crop. During validation and testing, we perform center cropping. The learning rate is set to 0.001 with an exponential decay rate of 0.98, and the model is optimized using the ADAM optimizer with a weight decay of 0.0005. We use a batch size of 8 and write our code in PyTorch [43]. All our experiments are carried out on a NVIDIA Titan 12G GPU.

### 5.1. Ablation studies

#### 5.1.1. The ablation study on the anchor balls

In this section, we build the ABRM using the VNet backbone through manual pruning. We compare the ABRM landmark detection loss with the baseline offsetmap regression loss. We do not perform the landmark classification task in this experiment, and learning is limited to the voxels within anchor balls. The settings of different anchor balls are listed in Table 3, with the  $r = 1u$  setting serving as the baseline for offsetmap regression. All networks are trained with the same hyperparameters for 2850 steps. Large-scale anchor balls involve many grid cells in network training and require more time for training. The learning curves for the training and validation sets are depicted in Fig. 5. We observe

Table 2

The statistics of landmarks on the training set.

|       | Cuspid | 2nd molar |     |     | 3rd molar |     |     |
|-------|--------|-----------|-----|-----|-----------|-----|-----|
|       | Cusp   | C         | R1  | R2  | C         | R1  | R2  |
| Left  | 400    | 404       | 404 | 384 | 404       | 403 | 342 |
| Right | 419    | 421       | 421 | 395 | 422       | 422 | 350 |

Table 3

The settings of anchor balls. The unit of each step's training time is second.

| No. | Anchor radiuses             | Training time (s/step) |
|-----|-----------------------------|------------------------|
| #1  | [ 1 ]                       | 1.43                   |
| #2  | [ 0.75, 1 ]                 | 2.44                   |
| #3  | [ 0.5, 0.75, 1 ]            | 3.69                   |
| #4  | [ 0.5, 0.75, 1, 1.25 ]      | 6.04                   |
| #5  | [ 0.5, 0.75, 1, 1.25, 1.5 ] | 14.91                  |

that more anchor balls lead to larger training regression errors but bring faster convergence on the validation set. The experimental results demonstrate that anchor balls can prevent the model from overfitting. However, increasing the radius of the anchor ball will increase the training time. For the remaining experiments, we choose the “#4” setting of anchor balls.

#### 5.1.2. The ablation study on landmark classification

We combine landmark detection and landmark classification because discriminating between existing and absent landmarks is critical in the 3D landmark detection task. To evaluate the landmark classification loss, we designed contrast experiments with or without hard negative mining and restriction loss terms. The receiver operator characteristic (ROC) curves on the validation and test sets in Fig. 6 depict our findings. The baseline classification loss achieved an area under the curve (AUC) of 0.9 and 0.904 in the validation and test sets, respectively. The hard negative mining loss term increased the AUC to 0.95 and 0.969 in the respective sets. Similarly, the restriction loss term resulted in an AUC of 0.928 and 0.93 in validation and test sets, respectively. The proposed integrated classification loss led to an AUC increase of 0.065 and 0.068, respectively. In conclusion, our experimental results suggest the effectiveness of the hard negative mining and restriction loss terms.

#### 5.1.3. The ablation study on network pruning

We constructed the network backbone by manually pruning upsampling blocks from popular encoder-decoder 3D-CNNs. To evaluate the effectiveness of the pruned network structures, we compared the offsetmap regression on the pruned networks with the heatmap regression on the source networks. The heatmap ground truth follows a Gaussian distribution with a variance of 0.1. Both the heatmap and offsetmap regression were conducted within a mask with a radius of 20 voxels. To mitigate the impact of absent landmarks, we built a complete subset of the dataset by removing CT volumes without all landmarks. This subset comprised 283, 56, and 60 volumes for training, validation, and testing, respectively. Our experimental results, shown in Table 4, indicate that the pruned network, which has similar parameters to the source

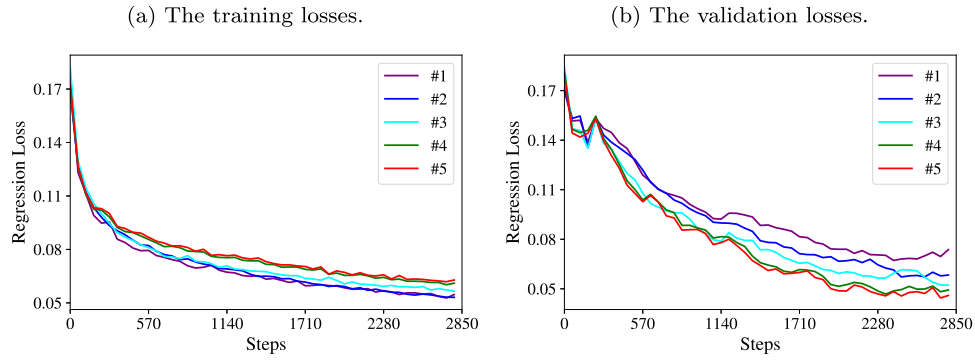


Fig. 5. The learning curves with different settings of anchor balls.

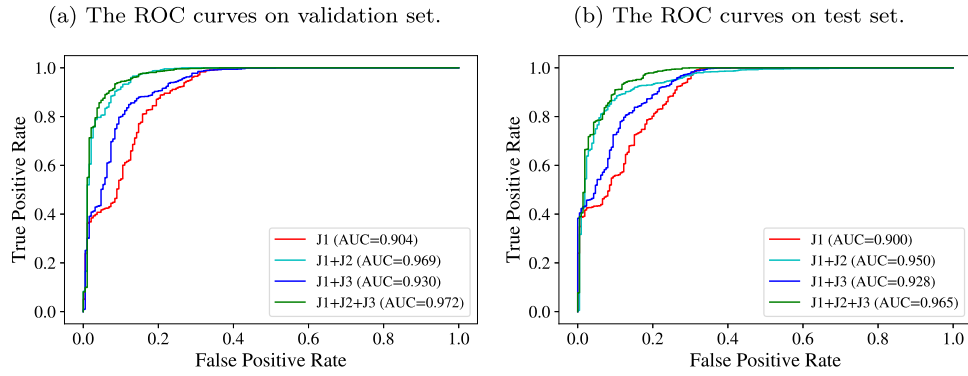


Fig. 6. The ROC curves with landmark classification settings. The “ $J\beta$ ,  $\beta \in \{1, 2, 3\}$ ” is corresponding to the term  $J_{cl}^{(\beta)}$  in Eq. (4), (5) and (6).

**Table 4**  
The MRE and SDR results on the data subset only with complete landmarks.

| Networks       | Datasets | Pruned | Params (M) | FLOPs (G) | MRE $\pm$ Std (mm)                | SDR (%)      |              |              |              |
|----------------|----------|--------|------------|-----------|-----------------------------------|--------------|--------------|--------------|--------------|
|                |          |        |            |           |                                   | 2 mm         | 2.5 mm       | 3 mm         | 4 mm         |
| VNet [37]      | Val      | ✓      | 44.93      | 157.1     | <b>1.94 <math>\pm</math> 0.67</b> | <b>66.43</b> | <b>77.98</b> | <b>85.83</b> | <b>93.93</b> |
|                |          | ✗      | 45.60      | 383.7     | 2.26 $\pm$ 1.56                   | 59.40        | 72.86        | 82.14        | 90.36        |
|                | Test     | ✓      | 44.93      | 157.1     | <b>1.99 <math>\pm</math> 0.71</b> | <b>67.86</b> | <b>78.31</b> | <b>85.20</b> | <b>91.20</b> |
|                |          | ✗      | 45.60      | 383.7     | 2.30 $\pm$ 1.17                   | 56.12        | 70.03        | 80.23        | 90.43        |
| UNet3D [38]    | Val      | ✓      | 3.57       | 63.6      | <b>1.98 <math>\pm</math> 0.41</b> | <b>67.86</b> | <b>80.98</b> | <b>85.00</b> | <b>92.31</b> |
|                |          | ✗      | 4.08       | 237.4     | 2.17 $\pm$ 0.61                   | 65.36        | 74.76        | 80.60        | 87.14        |
|                | Test     | ✓      | 3.57       | 63.6      | <b>1.90 <math>\pm</math> 0.65</b> | <b>65.94</b> | <b>77.81</b> | <b>86.99</b> | <b>93.54</b> |
|                |          | ✗      | 4.08       | 237.4     | 2.22 $\pm$ 0.74                   | 64.54        | 76.28        | 82.40        | 87.76        |
| ResUNet3D [39] | Val      | ✓      | 34.66      | 181.8     | <b>1.82 <math>\pm</math> 0.80</b> | <b>73.21</b> | <b>82.14</b> | <b>88.93</b> | <b>94.76</b> |
|                |          | ✗      | 35.32      | 399.1     | 2.05 $\pm$ 0.61                   | 58.45        | 75.83        | 85.36        | 93.93        |
|                | Test     | ✓      | 34.66      | 181.8     | <b>1.96 <math>\pm</math> 0.82</b> | <b>70.03</b> | <b>79.97</b> | <b>86.10</b> | <b>92.73</b> |
|                |          | ✗      | 35.32      | 399.1     | 2.09 $\pm$ 0.67                   | 58.67        | 71.81        | 82.65        | 93.11        |

network but dramatically reduces computational FLOPs, outperforms the heatmap regression on the source networks. The pruned versions of three popular encoder–decoder 3D-CNNs achieved an average decrease of 0.3 mm for MRE and an average increase of 9.76%, 7.12%, 4.93%, and 3.15% for SDR. These experimental results demonstrate that the offsetmap regression with the pruned networks performs better than the heatmap regression on the source networks.

#### 5.1.4. The ablation study on robustness

To evaluate the robustness of the ABRM, we compared it to the baseline offsetmap regression in detecting arbitrary landmarks. To this end, we created an incomplete dataset by omitting CT volumes with complete landmarks from the whole dataset. This incomplete subset comprised 175 samples for training, 44 for validation, and 40 for testing. Landmark detection was evaluated using MRE and SDR, while landmark classification was evaluated using accuracy and f1-measure. All models were built on the same pruned network backbone, but

ABRM involved the hybrid loss and anchor balls for network training. Experimental results on validation and test sets are presented in Table 5. The ABRM achieved an average 4.18 mm decrease in MRE and an average 57.10%, 66.05%, 66.79%, and 58.29% increase in SDR. In addition, ABRM achieved an average 5.59% and 6.05% increase in accuracy and f1-measure, respectively. In conclusion, the ABRM demonstrates more robust performance than the baseline offsetmap regression method for detecting arbitrary landmarks.

#### 5.2. Comparison on the whole MML dataset

Encoder–decoder networks can be utilized for landmark detection by replacing the segmentation loss with heatmap regression loss. Therefore, we compared the ABRM with baseline heatmap regression using three different network architectures: VNet, UNet3D, and ResUNet3D. We also compared the ABRM with Contour-Hugging Heatmaps (CHH) [44] using VNet. To adapt the CHH for the VNet, the loss function and

**Table 5**  
The MRE, SDR, accuracy, and F1-measure on the incomplete data subset.

| Networks       | Datasets | ABRM                     | MRE $\pm$ Std (mm)                                   | SDR (%)               |                       |                       |                       | Acc (%)               | F1 (%)                |
|----------------|----------|--------------------------|------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                |          |                          |                                                      | 2 mm                  | 2.5 mm                | 3 mm                  | 4 mm                  |                       |                       |
| VNet [37]      | Val      | $\times$<br>$\checkmark$ | 4.99 $\pm$ 1.47<br><b>2.48 <math>\pm</math> 1.28</b> | 11.88<br><b>54.21</b> | 23.27<br><b>70.05</b> | 30.94<br><b>77.97</b> | 47.77<br><b>87.62</b> | 90.05<br><b>91.40</b> | 90.79<br><b>93.74</b> |
|                | Test     | $\times$<br>$\checkmark$ | 5.62 $\pm$ 1.80<br><b>2.59 <math>\pm</math> 1.82</b> | 12.43<br><b>49.19</b> | 19.46<br><b>64.05</b> | 27.30<br><b>75.68</b> | 43.51<br><b>90.54</b> | 89.79<br><b>91.43</b> | 90.63<br><b>93.73</b> |
| UNet3D [38]    | Val      | $\times$<br>$\checkmark$ | 5.26 $\pm$ 1.85<br><b>2.21 <math>\pm</math> 1.46</b> | 12.62<br><b>61.63</b> | 20.05<br><b>78.96</b> | 27.72<br><b>86.39</b> | 45.54<br><b>90.59</b> | 86.85<br><b>92.69</b> | 89.07<br><b>94.71</b> |
|                | Test     | $\times$<br>$\checkmark$ | 6.21 $\pm$ 3.70<br><b>2.26 <math>\pm</math> 1.26</b> | 14.05<br><b>61.89</b> | 19.19<br><b>74.86</b> | 28.11<br><b>82.43</b> | 41.08<br><b>91.89</b> | 88.04<br><b>93.04</b> | 90.19<br><b>94.98</b> |
| ResUNet3D [39] | Val      | $\times$<br>$\checkmark$ | 5.81 $\pm$ 2.06<br><b>2.13 <math>\pm</math> 1.08</b> | 9.16<br><b>67.08</b>  | 14.85<br><b>77.72</b> | 21.78<br><b>84.90</b> | 40.35<br><b>92.82</b> | 86.04<br><b>92.86</b> | 88.19<br><b>94.82</b> |
|                | Test     | $\times$<br>$\checkmark$ | 6.98 $\pm$ 3.62<br><b>2.29 <math>\pm</math> 1.44</b> | 7.03<br><b>58.65</b>  | 11.35<br><b>72.79</b> | 18.65<br><b>81.08</b> | 33.24<br><b>89.46</b> | 84.64<br><b>91.96</b> | 87.05<br><b>94.18</b> |

**Table 6**  
The experimental results on the validation and test sets.

| Models             | Validation set                    |              |              |              |              | Test set                          |              |              |              |              |
|--------------------|-----------------------------------|--------------|--------------|--------------|--------------|-----------------------------------|--------------|--------------|--------------|--------------|
|                    | MRE $\pm$ Std (mm)                | SDR (%)      |              |              |              | MRE $\pm$ Std (mm)                | SDR (%)      |              |              |              |
|                    |                                   | 2 mm         | 2.5 mm       | 3 mm         | 4 mm         |                                   | 2 mm         | 2.5mm        | 3 mm         | 4 mm         |
| VNet [37]          | 3.23 $\pm$ 1.78                   | 39.42        | 54.96        | 66.20        | 78.35        | 3.00 $\pm$ 1.42                   | 41.50        | 55.39        | 65.91        | 79.12        |
| UNet3D [38]        | 2.54 $\pm$ 0.83                   | 45.29        | 59.59        | 70.91        | 86.36        | 2.80 $\pm$ 2.01                   | 44.11        | 57.74        | 70.62        | 85.77        |
| ResUNet3D [39]     | 2.74 $\pm$ 0.87                   | 38.26        | 53.55        | 66.69        | 84.55        | 3.00 $\pm$ 1.81                   | 37.04        | 51.09        | 64.81        | 82.41        |
| VNet with CHH [44] | 2.95 $\pm$ 1.33                   | 44.52        | 59.76        | 69.05        | 79.05        | 2.90 $\pm$ 1.56                   | 42.01        | 54.29        | 66.43        | 80.11        |
| Two-Stage [11]     | 1.78 $\pm$ 0.81                   | 68.93        | 82.48        | 88.68        | 95.37        | 1.92 $\pm$ 0.48                   | 69.11        | 82.32        | 89.73        | 94.53        |
| MTL [13]           | 1.91 $\pm$ 0.75                   | 69.70        | 80.47        | 87.29        | 93.86        | 1.97 $\pm$ 1.10                   | 68.10        | 81.16        | 86.94        | 92.98        |
| ABR-VNet           | 1.78 $\pm$ 0.80                   | 73.88        | 84.63        | 91.07        | 95.79        | 1.82 $\pm$ 0.73                   | 71.72        | 83.50        | 90.07        | 94.95        |
| ABR-UNet3D         | 1.74 $\pm$ 0.66                   | 75.00        | 85.02        | 90.82        | 96.04        | 1.77 $\pm$ 0.87                   | 73.97        | 85.21        | <b>91.98</b> | <b>96.45</b> |
| ABR-ResUNet3D      | <b>1.66 <math>\pm</math> 0.92</b> | <b>79.26</b> | <b>88.43</b> | <b>92.64</b> | <b>96.78</b> | <b>1.70 <math>\pm</math> 0.72</b> | <b>76.43</b> | <b>86.45</b> | 90.91        | 95.20        |

temperature scaling of the model are reproduced in 3D version following CHH's algorithms. Additionally, we performed the two-stage and MTL methods for comparison. The experimental results are presented in Table 6. Directly using heatmap regression on baseline models resulted in high MREs and low SDRs. However, the two-stage and MTL methods successfully limited the MRE to less than 2 mm. Among all methods, the ABRM achieved the best results, especially when using ResUNet3D as the network backbone.

### 5.3. Experimental analysis

To evaluate the performance of the compared methods on each landmark, we calculated the MRE of each landmark and illustrated them in Fig. 7, where “Cr”, “R1”, and “R2” indicate crown, root 1, and root 2, respectively. We observe three important observations. Firstly, detecting the roots of the third molar is more challenging than detecting those of the second molar due to the former's susceptibility to being impacted during growth. Secondly, crowns are easier to detect than roots because roots have a wide range of growth directions. Finally, root 2 is more difficult to be detected than root 1, possibly due to the fact that there are more root 2 than root 1. We have provided two examples with complete landmarks in Figs. 8 and 9, an example where the second molar only possesses one root in Fig. 10, and an example where the second and third molars were broken due to injury on the right mandibular in Fig. 11. All illustrated examples demonstrate the high performance of the ABRM.

## 6. Conclusion

The landmarks to be detected are arbitrary in the clinical need for 3D landmark detection. Thus, we focus on detecting arbitrary mandibular landmarks on 3D CT volumes. We release a large-scale benchmark dataset for mandibular molar landmarking (MML) in this

paper, which, to the best of our knowledge, is the largest dataset for 3D skull landmark detection. Inspired by YOLOV3, we performed an advanced ABRM, which combines landmark detection and landmark classification tasks for network training. The anchor ball module is designed to be responsible for generating candidate landmarks. A hybrid loss function is proposed to achieve the learning target. We also prune the popular encoder-decoder 3D-CNNs to obtain more efficient and lightweight network backbones. Overall, the ABRM outperforms the other methods on the incomplete data subset and whole dataset.

This paper addressed the genuine clinical necessity of data diversity in the CT volumes of various objects due to disease, injury or surgery. Through our experiments, we observed that detecting the roots of the third molar is more challenging compared to detecting the crowns, making it more difficult overall. We specifically focused on cases involving the development of teeth with either one or two roots, omitting data related to teeth with three roots as it is an exceptional scenario. To assess the reliability of detecting arbitrary landmarks, we measured the accuracy and f1-measure of landmark classification, aligning with the real clinical need. This method can be adapted for other 3D skull landmark detection tasks, like maxillofacial annotation and cephalometric annotation. Furthermore, it has the potential to be developed into a practical landmarking application that can assist clinical doctors in the future.

## 7. Discussion

Although the proposed ABRM significantly enhances the performance of 3D skull landmark detection, it is currently limited by two factors. Firstly, as discussed in Section 5.1.1, the proposed model inherits the limitation of anchor-based object detection from YOLOV3, wherein increasing the anchor ball radius results in higher computational complexity. To address this, we plan to optimize the ABRM by incorporating anchor-free object detection techniques, such as those



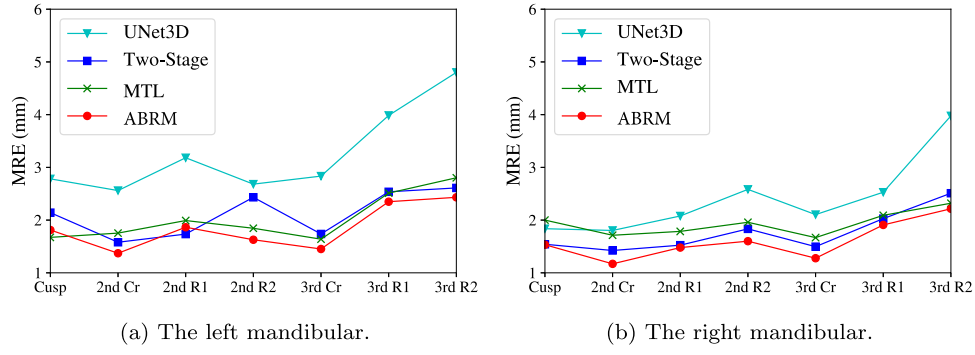


Fig. 7. The MREs of landmarks on the left and right mandibulars.

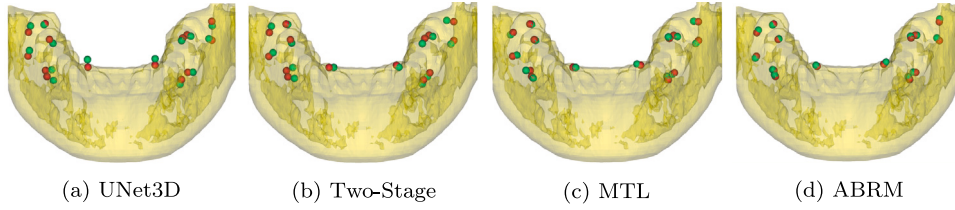


Fig. 8. Case 1. The CT volume has complete landmarks.

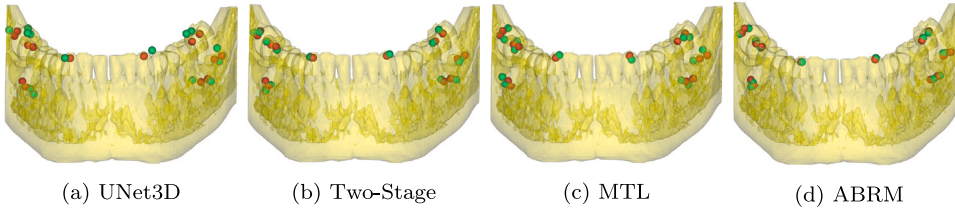


Fig. 9. Case 2. The CT volume has complete landmarks.

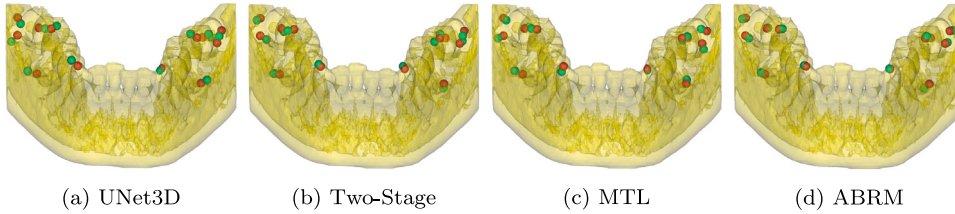


Fig. 10. Case 3. The 2nd molar has only one root.

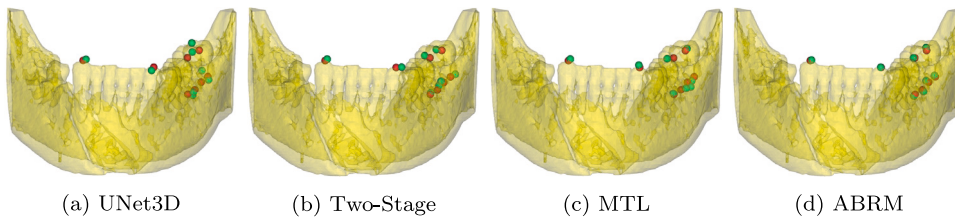


Fig. 11. Case 4. The 2nd and 3rd molars have been broken due to injury on the right mandibular.

presented in [45,46], in future work. Secondly, while the proposed model has been shown to effectively detect tooth landmarks, it has yet to be evaluated for detecting landmarks on the jawbone and skull in clinical 3D skull landmark detection. Moreover, since the proposed

model has not undergone evaluation in a multicenter study, its performance may vary depending on the CT volumes generated by different scanners. To address these limitations, we aim to conduct a multicenter study to assess the model's performance in detecting more landmarks.

## Ethical approval

Ethical Approval (Reference Number: WCHSIRB-D-2020-443) was obtained by the ethics committee of the West China Hospital. The informed consent was obtained from each subject. The personal privacy of subjects are protected because only the voxel array is available in our released CT volumes.

## CRediT authorship contribution statement

**Tao He:** Conceptualization, Methodology, Writing – original draft, Visualization. **Guikun Xu:** Software, Formal analysis, Data curation. **Li Cui:** Validation, Visualization. **Wei Tang:** Resources, Investigation. **Jie Long:** Supervision, Project administration. **Jixiang Guo:** Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The codes and dataset are accessible at [https://github.com/ithet10/07/mmld\\_code](https://github.com/ithet10/07/mmld_code).

## Acknowledgments

This work was supported by the National Natural Science Foundation of China under Grant 62206189, the National Natural Science Foundation of China under Grant 62376176, the China Postdoctoral Science Foundation under Grant 2023M732427 and the Special Fund for the Postdoctoral Research of Sichuan Province, China under Grant TB2022033, 2022 Dazhou City School Cooperation Special project, China under Grant 20226205.

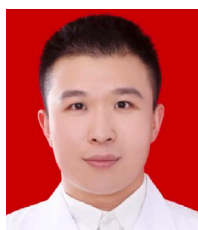
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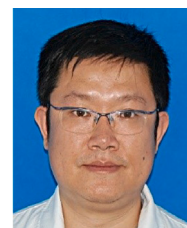
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